AJ MENDES

COMPUTER SCIENCE | CYBERSECURITY | QUANTUM ENGINEERING

Golden, CO | (720) 442-4724 | ajmendes17@gmail.com | Secret Security Clearance

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EDUCATION

Colorado School of Mines Expected May 2027

B.S. Computer Science | Minor in Military Science | Minor in Quantum Engineering | Cyber Defense Education Certificate | GPA: 3.5

RESEARCH EXPERIENCE

Undergraduate Researcher

AI Engine, Purdue University

- Participate in a team research project exploring satellite cybersecurity and anomaly identification through telemetry analysis.
- Study synthetic telemetry and simulated scenarios to learn normal satellite behavior and potential attack indicators.

Undergraduate Data Science Researcher

The Data Mine, Purdue University | CrowdStrike project | Jan 2026 - May 2026

- Built Python tools to evaluate K-means clustering on labeled synthetic cybersecurity data using accuracy, adjusted Rand index, and normalized mutual information.
- Tested noise, cluster-count, and feature variations; observed approximately 98% cluster-assignment accuracy in labeled tests with up to 20 clusters.
- Developed centroid-matching and drift visualizations with PCA, UMAP, and t-SNE; presented findings to Purdue, Data Mine of the Rockies, and CrowdStrike collaborators.

PROJECTS

Local Learning Studio

React | TypeScript | Fastify | SQLite | Codex CLI

- Exports source-linked course notes and progress to an Obsidian knowledge base.
- Tracks saved study sessions and quiz attempts with transparent practice progress.
- Uses authenticated Codex CLI workflows, avoiding separate pay-as-you-go API setup by default.

Secure Chat Application

Python | RSA-OAEP | AES-256-GCM | TCP

- Built a TCP client-server chat application with RSA-2048/OAEP key exchange and AES-256-GCM authenticated encryption; the relay forwards ciphertext without access to plaintext.

MarketPulse AI

Python | pandas | scikit-learn | TensorFlow/Keras | Streamlit

- Built an ML research pipeline with return, volatility, moving-average, volume, RSI, and MACD features for seven equities benchmarked against SPY.
- Compared logistic regression, random forest, and neural-network models using time-aware backtests, transaction costs, threshold sweeps, automated reports, and a Streamlit dashboard.

ADDITIONAL EXPERIENCE

AI Research Contributor 2026 - Present

Handshake AI

- Create and source-verify complex research tasks to evaluate advanced AI systems; analyze responses for factual accuracy, research quality, and failure patterns.

Electrical Control Panel Technician Jun 2025 - Aug 2025

Xpect Solutions | Longmont, CO

- Assembled and wired industrial electrical control panels from schematics, wiring diagrams, and technical drawings.

Construction and General Labor Summers 2020 - 2024

Formit Right Concrete | Longmont, CO

- Completed five summers of full-time concrete, site-preparation, material-handling, and landscaping work.

TECHNICAL SKILLS

Languages: Python, SQL, R, Java, C++, TypeScript/JavaScript
Data and ML: pandas, NumPy, scikit-learn, TensorFlow/Keras, PCA, UMAP, t-SNE, clustering, backtesting
Tools: Git, Linux, Streamlit, Matplotlib

LEADERSHIP & CREDENTIALS

Army ROTC, Colorado School of Mines | Former Cadet Company Commander; currently Assistant Operations Officer (AS3)